

# Munkres 1

Vincent Wang

April 2025

## 1 Introduction

Finite sets. Let  $f : A \rightarrow \{1, 2, 3, \dots, n\}$  be a bijection and  $B \subsetneq A$ . Then  $B = \emptyset$  or there exists bijective  $g : B \rightarrow \{1, 2, 3, \dots, m\}$  for some  $m < n$  and there doesn't exist a bijection  $B$  to  $\{1, 2, 3, \dots, n\}$ .

We proceed by induction on  $n$ . If  $n = 1$ , the result holds. Let  $c \in A - B$ . There exists a bijection  $f_1 : c \mapsto n + 1, A - c \rightarrow [n]$ . If  $B = A - c$ , we're done. Else  $B \subsetneq A - c$ . By IH,  $f_2 : B \rightarrow [k]$ , where  $k < n < n + 1$ .

Suppose  $m < n$ . Then there is no bijection  $[m] \leftrightarrow [n]$ . We induct on  $n$ .  $n = 1$  holds.  $m \neq 0$ . Pick  $f : m \mapsto n, [m - 1] \rightarrow [n - 1]$ . By IH, no bijection exists.

Since  $A$  is finite,  $A \leftrightarrow [n]$  for some  $n \in \mathbb{N}$ . Since  $B \subset A$ ,  $B \leftrightarrow [m]$  for some  $m \leq n$  or  $B = \emptyset$ , so  $B$  is finite. If in particular  $B \subsetneq A$ , then  $m < n$  ( $m = 0$  when  $B = \emptyset$ ).

We show (1)  $\rightarrow$  (2), (2)  $\rightarrow$  (3), (3)  $\rightarrow$  (1). If  $B$  is finite,  $B \leftrightarrow [n]$ , so  $f : [n] \rightarrow B$  is surjective. If  $[n] \twoheadrightarrow B$ , then  $B \hookrightarrow [n]$ , so  $B \leftrightarrow [m]$  for some  $m \leq n$ .  $B$  is finite.

Check if  $A, B$  are empty. Suppose  $A, B$  are finite. Suppose  $A \leftrightarrow [m]$  and  $B \leftrightarrow [n]$ . Then  $[m + n] \twoheadrightarrow A \cup B$ , so  $A \cup B$  is finite.  $A \times B \leftrightarrow [m] \times [n]$  and  $[m] \times [n] \leftrightarrow [mn]$  by  $(r, c) \mapsto n(r - 1) + (c - 1)$ .

If  $A, B \neq \emptyset$  and  $A \times B$  is finite, then so are  $A, B$ .  $A \leftrightarrow A \times \{b\} \subset A \times B$ .

$A, B$  are finite, so  $A \times B$  is finite.  $B^A \subset P(A \times B)$ . Since  $A \times B$  is finite, then so is  $P(A \times B)$ .

Let  $\mathcal{A}$  be a countable collection of countable sets indexed by  $\mathbb{N}$ . Denote by  $f(n)$  the set  $A \in \mathcal{A}$  associated with  $n$ . Each  $A$  is indexed by  $\mathbb{N}$ . Denote by  $f(n, m)$  the  $m$ th element of the  $n$ th element of  $\mathcal{A}$ . Let  $U$  denote  $\bigcup \mathcal{A}$ .  $f : \mathbb{N} \times \mathbb{N} \twoheadrightarrow U$ . Since  $\mathbb{N} \leftrightarrow \mathbb{N} \times \mathbb{N}$ ,  $\mathbb{N} \twoheadrightarrow U$ , so  $U$  is countable.

Let  $X = \{0, 1\}$ . Let  $\mathcal{B}$  denote the set of countable subsets of  $X^\omega$ . Show that  $\mathcal{B}$  and  $X^\omega$  have the same cardinality.

By AOC, there exists a function  $f : \mathcal{B} \rightarrow \mathcal{C}$ , where  $\mathcal{C}$  is the set of all sequences of sequences in  $X$ . There exists  $g : \mathcal{C} \rightarrow X^\omega$  since: (1) suppose  $h : N \times N \leftrightarrow N$ , and  $C \in \mathcal{C}$ . (2) Then  $g(C)(n) = C(h(n))$  determines a unique sequence in  $X^\omega$ .  $g \circ f : \mathcal{B} \rightarrow X^\omega$  is bijective.

Axiom of Choice. Let  $\mathcal{B}$  be a collection of nonempty sets. There exists a function  $c : \mathcal{B} \rightarrow \cup \mathcal{B}$  such that  $c(B) \in B$  for all  $B \in \mathcal{B}$ . Proof: For each  $B \in \mathcal{B}$  define  $B' = \{(B, b) : b \in B\}$ . Let  $A = \{(B, b) \in \mathcal{B} \times \cup \mathcal{B} : B \in \mathcal{B} \wedge b \in B\}$ . For fixed  $B$  denote by  $B' = \{(B, b) : b \in B\}$ . The set of possible  $B'$  partition  $A$  since if  $(B, b) = (D, d)$ , then  $B = D$ , so  $B' = D'$ . By AOC, there exists a set  $C \subset A$  such that  $C \cap B' = \{(B, b)\}$  for all  $B \in \mathcal{B}$ .  $C$  is a function since if  $(B, b), (B, b') \in C$  for  $b \neq b'$ , then  $C \cap B' = \{(B, b), (B, b')\}$ , contradiction.  $C$  has domain  $\mathcal{B}$ , so we have found our choice function  $c : \mathcal{B} \rightarrow \cup \mathcal{B}$ .

Let  $A$  be an infinite set. There exists an injection  $f : N \rightarrow A$ . Let  $c : P(A) - \{\emptyset\} \rightarrow A$  be a choice function. Let  $f(1) \in A$  be arbitrary. Define  $f(n+1) = c(A - f([n]))$ . Since  $A$  is infinite,  $A - f([n]) \neq \emptyset$  as otherwise  $f$  would be a surjection  $f : [n] \twoheadrightarrow A$ .  $f$  is well-defined by the Induction principle. \*Pick  $f(1) = c(A)$  to avoid arbitrary choices.

Let  $\{f_n\}_{n \geq 1}$  be a family of  $N$ -indexed injective functions  $f_n : [n] \hookrightarrow A$ . Show that  $A$  is infinite. Define  $F : N \rightarrow A$  by  $F(1) = f_1(1)$  and  $F(n+1) = f_{n+1}(m)$ , where  $m$  is the smallest  $m \in [n+1]$  such that  $f_{n+1}(m) \notin F([n])$ .  $F(n+1)$  is well-defined because  $f_{n+1}([n+1])$  has cardinality  $n+1$  and  $F([n])$  has cardinality  $n$ , so  $f_{n+1}([n+1]) \not\subset F([n])$ .  $F : N \hookrightarrow A$ , so  $A$  is infinite.

Well Ordering theorem implies AOC. Let  $\mathcal{A}$  be a collection of nonempty disjoint sets. Well order  $\bigcup \mathcal{A}$  and let  $C = \{\min(A) : A \in \mathcal{A}\}$ . Note that  $C \cap A = \{\min(A)\}$ , a singleton element.

Let  $S_\Omega$  denote the smallest uncountable Well-Ordered set. Let  $X_0$  denote the subset of  $S_\Omega$  of elements with no immediate predecessor. Show  $X_0$  is uncountable.

Let  $J$  be a well ordered set. A subset  $J_0 \subset J$  is said to be inductive if for all  $\alpha \in J$ ,  $J(\alpha) \subset J_0 \rightarrow \alpha \in J_0$ . Show that  $J_0 = J$ .

Sps FTSOC  $J_0 \subsetneq J$ . Let  $\alpha \in J - J_0$  be minimal. Then  $J(\alpha) \subset J_0$ , so  $\alpha \in J_0$ , contradiction. It follows  $J_0 = J$ .

Let  $\mathcal{A}_W$  be a collection of well ordered sets  $A$  indexed by a well ordered set  $W$ . Suppose  $x, y \in \bigcup \mathcal{A}$ . If  $x, y$  are in the same well ordered set  $A$ , compare  $x, y$  with the order of  $A$ . Else, compare the indices of the well ordered sets

associated with  $x, y$ . This is a generalization of the product of two well ordered sets with dictionary order.

Let  $W$  be a well ordered set and  $A \subset W$  st  $A \neq \emptyset$ . Let  $A$  be unbounded, so that there does not exist  $b \in W$  st  $b \geq A$ . Show that  $B = \bigcup_{\alpha \in A} W(\alpha) = W$ .

Let  $w \in W$  be arbitrary. There exists  $\alpha \in A$  st  $\alpha > w$ , so  $w \in W(\alpha)$ , so  $w \in B$ . That is,  $W \subset B$ . However,  $B \subset W$ , so  $B = W$ .

Suppose  $A$  is bounded. Then  $B = W(\sup(A))$ . If  $w \in B$ , then  $w < \alpha$  for some  $\alpha \in A$ , so  $w < \sup(A)$  as  $\alpha \leq \sup(A)$ .  $B \subset W(\sup(A))$ . Conversely, if  $w \in W(\sup(A))$ , then  $w < \sup(A)$ , so  $w < \alpha$  for some  $\alpha \in A$ . Hence,  $w \in B$  and  $W(\sup(A)) \subset B$ .  $W(\sup(A)) = B$ .

Let  $\Omega$  be the minimal uncountable well ordered set. Denote by  $X_0 \subset \Omega$  the set of elements with no immediate predecessor. Then  $X_0$  is uncountable.

Let  $w \in \Omega$ . Since  $\Omega$  has no largest element, there exists  $w' > w$ . Further, there exists an immediate successor  $w^+$  of  $w$ . Consider the set  $W$  of immediate predecessors of  $w$  in sequence:  $w^+, w^{++}, w^{+++}, \dots$ . If  $W$  is unbounded, then  $\bigcup_{w \in W} \Omega(w) = \Omega$ , a contradiction since the LHS is a countable union of countable sets. Hence,  $W$  is bounded and  $\alpha = \sup(W)$  is well defined.  $W$  has no maximum since if  $s \in W$ , then  $s^+ \in W$ . It follows that  $\alpha$  has no immediate predecessor, so  $\alpha \in X_0$ . By construction,  $\alpha > w$ , so  $X_0$  is indeed unbounded.

We have  $\bigcup_{x \in X_0} \Omega(x) = \Omega$ . Since each section  $\Omega(x)$  is countable,  $X_0$  must be uncountable.

Let  $W$  be a well ordered set and  $\mathcal{F}$  be the set of all functions from sections of  $W$  to a set  $C$ . Denote by  $\rho : \mathcal{F} \rightarrow C$  a given function. Then there exists a unique function  $h : W \rightarrow C$  such that  $h(x) = \rho(h|W(x))$  (1) for all  $x \in W$ .

Suppose  $h_1 : W(x_1) \rightarrow C$  and  $h_2 : W(x_2) \rightarrow C$  satisfy (1). Sps FTSOC  $h_1(x) \neq h_2(x)$  for some  $x < x_1, x_2$ . Let  $m$  be the smallest such  $x$ , so  $h_1|W(m) = h_2|W(m)$ . Hence,

$$h_1(m) = \rho(h_1|W(m)) = \rho(h_2|W(m)) = h_2(m),$$

contradiction. (2).

Let  $W_0 = \{w \in W : \text{there exists a function } h : W[w] \rightarrow C \text{ such that (1)}\}$ . Suppose  $W(\alpha) \subset W_0$ . If  $\alpha$  has an immediate predecessor  $\alpha^-$ , there is a function  $h^- : W[\alpha^-] \rightarrow C$  satisfying (1). By construction  $W[\alpha^-] = W(\alpha)$ . We can extend  $h^-$  to  $h = h^- \cap (\alpha, \rho(h^-))$ , which satisfies (1), so  $\alpha \in W_0$ . Else,  $\alpha$  has no predecessor, so  $W(\alpha) = \bigcup_{\beta < \alpha} W(\beta)$ . Consider  $H = \bigcup_{\beta < \alpha} H_\beta$ , where capital  $H$  denotes the subset treatment of a function  $h$ .  $H$  is a function: if  $(w, c), (w, c') \in H$ , there exist  $\beta, \beta' < \alpha$  such that  $(w, c) \in H_\beta$  and  $(w, c') \in H_{\beta'}$ .

$w$  is a common point in the domains of  $h_\beta, h_{\beta'}$  and  $h_\beta, h_{\beta'}$  satisfy (1) by assumption, so by (2),  $c = c'$ . The domain of  $H$  is  $W(\alpha)$ . Finally, if  $w \in W(\alpha)$ , there exists  $\beta$  such that  $w < \beta < \alpha$ , so

$$h(w) = h_\beta(w) = \rho(h_\beta|W(w)) = \rho(h|W(w)).$$

We can extend  $H : W(\alpha) \rightarrow C$  to  $K : W[\alpha] \rightarrow C$  by taking  $K(\alpha) = \rho(H|W(\alpha)) = \rho(H)$ , so  $K$  satisfies (1). Hence,  $\alpha \in W_0$ , so  $W_0$  is an inductive subset of  $W$  and  $W_0 = W$ . (3)

Define  $h : W \rightarrow C$  by  $h(w) = k(w)$  for any  $k$  from (3).  $h$  is well defined from (1). Further,  $h$  also satisfies (1) and is unique.

Let  $J, K$  be well ordered sets and  $h : J \rightarrow K$ . Then the following are equivalent:

- (a)  $h$  is order preserving and its image is  $K$  or a section of  $K$ .
- (b)  $h(\alpha) = \min(K - h(J(\alpha)))$ .

Consider  $y \in h(J(\alpha))$ . Then  $y = h(x)$  for some  $x < \alpha$ . Since  $h$  is increasing,  $y < h(\alpha)$ , so  $y \in K(h(\alpha))$ . Conversely, if  $y \in K(h(\alpha))$ , then  $y < h(\alpha)$ . Since  $h(\alpha)$  lies in the image of  $h$  and  $y < h(\alpha)$ ,  $y$  also lies in the image of  $h$ . That is, there exists  $x \in J$  such that  $y = h(x)$ . It follows  $x < \alpha$ , so  $y \in h(J(\alpha))$ . Hence,  $h(J(\alpha)) = K(h(\alpha))$ .

$$\min(K - h(J(\alpha))) = \min(K - K(h(\alpha))) = \min\{k \in K : k \geq h(\alpha)\} = h(\alpha).$$

Conversely, assume  $h(\alpha) = \min(K - h(J(\alpha)))$ . If  $\beta < \alpha$ , then  $J(\beta) \subset J(\alpha)$ , so  $h(J(\beta)) \subset h(J(\alpha))$ , so  $K - h(J(\alpha)) \subset K - h(J(\beta))$ , so  $h(\alpha) \geq h(\beta)$ .  $\beta \in J(\alpha)$ , so  $h(\beta) \in h(J(\alpha))$ , so  $h(\beta) \notin K - h(J(\alpha))$ . Hence,  $h(\alpha) \neq h(\beta)$ , so  $h$  is strictly increasing. Suppose  $k$  lies in the image of  $h$  and  $k' < k$ . Let  $k = h(j)$ , so  $k = \min(K - h(J(j)))$ . Then  $k' \in h(J(j))$ , so  $k'$  also lies in the image of  $h$ . The image of  $h$  is either a segment of  $K$  or all of  $K$ .

1.